

ISHIKA DUTTA

Software Engineer | Full-Stack Developer | AI/ML Researcher | Tech Risk Enthusiast

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B.Tech CS Engineering student (CGPA: 8.5) | 1+ year internship experience in full-stack dev, AI/ML research, cloud & technology risk — currently Technology Risk Intern @ EY India. Published IEEE researcher; Infosys Global Hackathon State Winner & Top 10 Global Finalist (650+ teams). Proficient in Python, Java, MERN stack, AWS, Docker, Kubernetes, CI/CD, machine learning, LLM fine-tuning, RAG pipelines, vector databases, system design, unit testing & Agile/Scrum.

TECHNICAL SKILLS

Languages: Python, Java, C, JavaScript, ANSI SQL

Web Dev: React.js, Node.js, Express.js, Next.js, HTML5, CSS3, REST APIs

Databases: MongoDB, PostgreSQL, MySQL, Prisma ORM, Supabase

AI / ML: TensorFlow, PyTorch, Deep Learning, Generative AI, Agentic AI, Federated Learning, NLP, LLM Fine-Tuning, RAG, Vector DBs, Quantum ML

Cloud & DevOps: AWS (EC2, S3, Lambda), Docker, Kubernetes, CI/CD Pipelines, Microservices, Git, GitHub

Tools: FastAPI, Pinecone, Postman, Packet Tracer, Arduino

Core CS: DSA, OOP, OS, DBMS, Cybersecurity, Computer Networks, System Design, Unit Testing, Agile/Scrum

Tech Risk: ITGC, ITAC, SAP, SOC, SOX, COBIT, ISO 27001

PROFESSIONAL EXPERIENCE

Technology Risk Intern

EY India

- Working across ITGC, ITAC, SAP & SOC engagements — IT general controls and application controls testing, and SOC reporting for enterprise clients.

May 2026 – Jul 2026

Software Engineering Intern

iQuster Solutions LLP, Kolkata

- Engineered an AI-powered learning platform (React.js, Node.js, MongoDB) for 100+ students; optimised API calls to cut page load time by ~28%.
- Architected beginner AI/ML curriculum modules, spearheading a ~35% improvement in learning-path completion rate.

Dec 2025 – Mar 2026

Student Co-Lead (Research)

IEM UEM Generative AI CoE, Kolkata

- Orchestrated 8+ students across LLM fine-tuning & RAG pipeline experiments; cut research turnaround by ~20% with Agile sprint workflows.
- Collaborated with 3 faculty mentors to present findings at 2 international conferences; contributed to IEEE publications.

Jul 2025 – Ongoing

Software Engineering Intern – AI/ML Research

IEEE Pune Section

- Co-authored IEEE ICDSINC 2025 paper; benchmarked 3 machine learning model variants achieving F1 score of 0.87 on ICU event prediction.

Jun 2025 – Aug 2025

Full-Stack Software Engineering Intern

Generative AI CoE

- Architected full-stack AI trading simulator (React.js, Flask, Node.js); integrated WebSocket feeds reducing data latency by ~40%.

May 2025 – Jul 2025

Full-Stack Software Engineering Intern

Fluttr

- Engineered 6+ web features across 2 Agile sprints; spearheaded MongoDB indexing strategy improving API response time by ~25%.

May 2025 – Jun 2025

Campus Ambassador

Kshitij, IIT Kharagpur

- Represented IIT Kharagpur's Kshitij fest to 3,000+ students; orchestrated outreach driving a 15% increase in campus registrations.

Nov 2025 – Jan 2026

AWS Cloud Engineering Intern

NSIC Technical Services Centre (Govt. of India)

- Provisioned 3 cloud apps on EC2, S3, Lambda; engineered IAM role automation reducing manual deployment steps by ~50%.

Sep 2024 – Oct 2024

Research Engineering Intern

mjunction services ltd

- Authored 25+ page Odoo ERP technical report adopted for internal vendor evaluation.

Jun 2024 – Jul 2024

Project Intern

Nest Innovative Solutions Pvt. Ltd.

- Architected an IoT RFID door lock system using Arduino; achieved ~98% access accuracy across 200+ hardware-software integration test cycles.

May 2024 – Jun 2024

PROJECTS

ITGC Controls Toolkit | [GitHub](#) | Python, Flask, SQLite, RBAC

- Built an audit-grade toolkit testing IT general controls end to end — SoD analysis, change auditing, access recertification & a control map across SOX, COBIT, ISO 27001, NIST CSF & SOC 2.

Intelligent Talent Acquisition Assistant | [GitHub](#) | *Python, LangChain, Agentic AI, Next.js, RAG*

- Architected a multi-agent AI recruitment pipeline automating resume screening, candidate ranking & interview scheduling; cut manual screening by ~60%.

Equity Technology Risk Lab | [GitHub](#) | *Python, Stock Prediction, Risk Analysis*

- Developed a stock-prediction dashboard with built-in risk analysis — the model-risk companion to the ITGC toolkit.

Emotion Detection System | *Python, Bidirectional LSTM, NLP, TensorFlow*

- Engineered a sentiment/emotion classifier using Bidirectional LSTM achieving 89% accuracy on real-time text analysis.

Foul Comment Rectification System | [GitHub](#) | *Python, LLM Fine-Tuning, NLP, Sentiment Analysis*

- Spearheaded an LLM-driven toxicity filtering & rephrasing pipeline; reduced toxic content pass-through by ~75% in moderation tests.

FarmConnect – Full-Stack Farmer Platform | [GitHub](#) | *React.js, Node.js, Twilio, Multilingual NLP*

- Orchestrated a full-stack platform with voice assistance, multilingual NLP, Twilio SMS & insurance advisory; removed language/accessibility barriers for farmers.

GeoGrow · KisanSure | [GitHub](#) | *TypeScript, React, Vercel*

- Shipped a live AI guardian for farmers — crop-insurance guidance and agri-intelligence, deployed on Vercel.

AirCanvas | [GitHub](#) | *Python, OpenCV, MediaPipe, NumPy*

- Engineered a computer vision application enabling real-time air drawing using hand gesture recognition via MediaPipe and OpenCV.

AI EdTech Platform | [GitHub](#) | *React.js, Node.js, Supabase, PostgreSQL*

- Architected an EdTech platform for school students with personalised AI/ML learning paths; leveraged Supabase and PostgreSQL for scalable data management.

SaathiBazaar | [GitHub](#) | *JavaScript, CSS, HTML*

- Architected a community marketplace web app connecting local buyers and sellers.

AI Trading Simulator | [GitHub](#) | *React.js, Node.js, Flask*

- Engineered a full-stack AI-powered trading portfolio simulator with real-time market simulation features.

EDUCATION

B.Tech CSE — IEM, Kolkata | 2022–2026 | CGPA: 8.5
ISC Class XII — La Martiniere for Girls, Kolkata | 82%
ICSE Class X — La Martiniere for Girls, Kolkata | 89%

CERTIFICATIONS

- Neural Networks & Deep Learning | DeepLearning.AI | Sep 2025
- AWS Cloud | NSIC | Credential: 217157
- EDA for ML | IBM | Nov 2024 | Credential: RNOTIZ5EV2TT
- Intro to Data Analytics | IBM | Apr 2025

RESEARCH PUBLICATIONS

- Federated Transformer-Based Learning for Predicting ICU Events with Multi-hop XAI Explanation Models — IEEE ICDSINC (2025)
- Enhancing Remote Sensing Image Classification with Hybrid Quantum-Classical Neural Network — IRTM Conference (2025)
- Astrophysics Research Paper — IEM Physics International Conference; Featured in IEM Magazine

ACHIEVEMENTS & AWARDS

- State Winner & Top 10 Global Finalist — Infosys Global Hackathon 2025 (Team Leader, 650+ teams; Hyderabad Finals)
- Best Session Paper Award — ICELTS International Conference
- Smart India Hackathon (SIH) 2024 — Internal Hackathon Participant
- Delegate — Summit Sikkim 8.0 MUN (UNCSW)
- Campus Ambassador — Kshitij, IIT Kharagpur

LEADERSHIP & EXTRACURRICULAR

- PR Lead — IEEE CS IEM Student Branch (2025–26): Public relations, outreach & branding.
- Research Co-Lead — IEM GenAI CoE: Student research coordination & IEEE publication support.
- Event Coordinator — Converge | IEEE CS (Mar 2025): Event planning & online sessions.
- Coordinator — COD Event, IEMPACT Fest (Feb 2025): Introduced new event segment.
- Organising Committee — IEM MUN 2024: Large-scale academic conference execution.